

0.前置

$$\Pi_0 = On$$

$$\forall \alpha \in On \forall A (\forall \beta \exists \gamma (\beta, \gamma \in A \leftrightarrow \beta \in \gamma)) \leftrightarrow A \text{ aft } \alpha = \{\mu : \alpha \in \mu, \mu \in A\}$$

$$A = (a_1, \dots, a_n) (a_i \in On, 1 \leq i \leq n):$$

$$\bullet (n = 1 \rightarrow B = \Pi_{a_1}) \vee (n > 1 \wedge ((a_1 \leq a_2 \rightarrow B = \Pi_{a_n} \text{ onto } str((a_2, \dots, a_n))) \vee (a_1 > a_2 \rightarrow B = \Pi_{a_n} \cap str((a_2, \dots, a_n)))) \leftrightarrow str(A) = B$$

$$\bullet gp(A) = \begin{cases} (a_1, \dots, a_{n-1}) & a_n \leq 1 \\ (a_{i+1}, \dots, a_n) & \exists i(i \in \{1, \dots, n\} \wedge a_i < a_1 \wedge \neg \exists j(j \in i \wedge a_j < a_1)) \\ (a_2, \dots, a_n) & \neg \exists i(i \in \{1, \dots, n\} \wedge a_i < a_1 \wedge \neg \exists j(j \in i \wedge a_j < a_1)) \end{cases}$$

$$\bullet bp(A) = \begin{cases} () & a_n \leq 1 \\ (a_1 - 1) & n \leq 1 \wedge a_1 > 1 \\ (a_2, \dots, a_i, a_1 - 1) & \exists i(i \in \{1, \dots, n\} \wedge a_i < a_1 \wedge \neg \exists j(j \in i \wedge a_j < a_1)) \\ (a_2, \dots, a_n, a_1 - 1) & \neg \exists i(i \in \{1, \dots, n\} \wedge a_i < a_1 \wedge \neg \exists j(j \in i \wedge a_j < a_1)) \end{cases}$$

$$(A \text{ onto})^\alpha B = \begin{cases} A \text{ onto } (A \text{ onto})^\beta B & \alpha = \beta + 1 \\ \{\mu \mid \forall \beta < \alpha, \mu \in (A \text{ onto})^\beta B\} & \end{cases}$$

$$\alpha \text{ th } A = \begin{cases} \sup\{\nu : \mu \text{ th } A, \mu \in \alpha\} & \alpha > 0 \\ 1 & \alpha = 0 \end{cases}$$

1. BOCF

$$C(\alpha, \beta, 0) = \beta \cup \{\min \Pi_i : 1 < i < \omega\}$$

$$C(\alpha, \beta, n+1) = \{\nu + \mu, \psi_\xi(\gamma) : \gamma, \nu, \mu, \xi \in C(\alpha, \beta, n) \wedge \gamma \in \alpha \cap C(\gamma, \xi)\}$$

$$C(\alpha, \beta) = \bigcup_{i \in \omega} C(\alpha, \beta, i)$$

$$\psi_\pi(\alpha) =$$

$$\begin{cases} \min\{\nu : \nu \in (str(bp(B)) \text{ onto } str(gp(B)))/C(\alpha, \xi \text{ th } A)\} & \pi = \xi + 1 \text{ th } A \wedge \xi > 0 \wedge \exists B(A = str(B)) \\ 0 & \pi = 0 \end{cases}$$